MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) FIFTH SEMESTER EXAMINATION

(2024 ADMISSION ONWARDS)

MG5DSESTA302– STATISTICAL COMPUTING USING PYTHON

Duration: 1Hour Maximum: 30 Marks

PRACTICAL EXAMINATION

Answer any five questions. Each question carries 6 marks.

Solve the questions using Jupyter notebook in **the computer lab. The python codes along with output must be written in the answer sheets.**

1. Load the iris data set from sklearn.datasets. Standardize the four features using StandardScaler. Apply PCA with 2 components. Print the explained variance ratio of each component and the cumulative explained variance. Plot the first two principal component scores as a scatter plot coloured by the species. Write the codes used and output obtained. [A][4]

2. The table below gives a distance matrix of perceived dissimilarities between 5 air freshener brands measured during a market study. Conduct Classical Metric Multidimensional Scaling on this precomputed distance matrix with n_components=2, and random_state=42. [A][4]

Brand	**Godrej**	**Airwick**	**Odonil**	**Ambi Pure**	**Lia**
Godrej	0	206	963	2976	1949
Airwick	206	0	802	2815	1771
Odonil	963	802	0	2013	996
Ambi Pure	2976	2815	2013	0	1307
Lia	1949	1771	996	1307	0

3. Load the diabetes dataset from sklearn. Standardize the 10 predictor variables and fit factor analysis with n_components=3. Print the first five factor scores. [A][4]

4. Using the fetch_california_housing dataset available in sklearn.datasets, fit a multiple linear regression model using all predictor variables. Split the data using test_size=0.20 and random_state=501, and evaluate the model using MSE, RMSE, MAE and R^2. [A][5]

5. Train a logistic regression model on the load_breast_cancer dataset. Use 80% of the sample as training data and split with random_state = 42. Use max_iter = 100. Measure the accuracy using accuracy_score print the confusion matrix and display the confusion matrix. Write the codes used and the output obtained. [A][5]

6. Load the load_iris dataset from sklearn.datasets. Split the data into training and testing sets in the ratio 80:20 using random_state=101. Standardize the features using StandardScaler and train an SVM classifier using an RBF kernel with C=1 and gamma= 'scale'; Report the classification accuracy and predict the class of a new flower having the following measurements: sepal length = 5.8 cm, sepal width = 2.7 cm, petal length = 4.1cm, and petal width = 1.0 cm. [A][5]

7. Train a decision tree classifier using data set load_iris from sklearn.datasets. Using 75% of the sample as training data. Split the data with random_state = 88. Use gini as criterion, max_depth = 3 and random_state =11. Print the test accuracy and confusion matrix. Also predict the species of a flower with measurements [5.9, 3.0, 5.1, 1.8]. Write the codes used and output obtained. [A][5]

8. Using the Iris dataset, apply K-Means to the four measurements. Use the Elbow Method to choose k, then form k clusters with random_state=404. Display the cluster labels and centroids, and plot petal length against petal width. [A][5]